

02464 Artificial Intelligence and Human Cognition

Visual Perception as Bayesian Inference 2

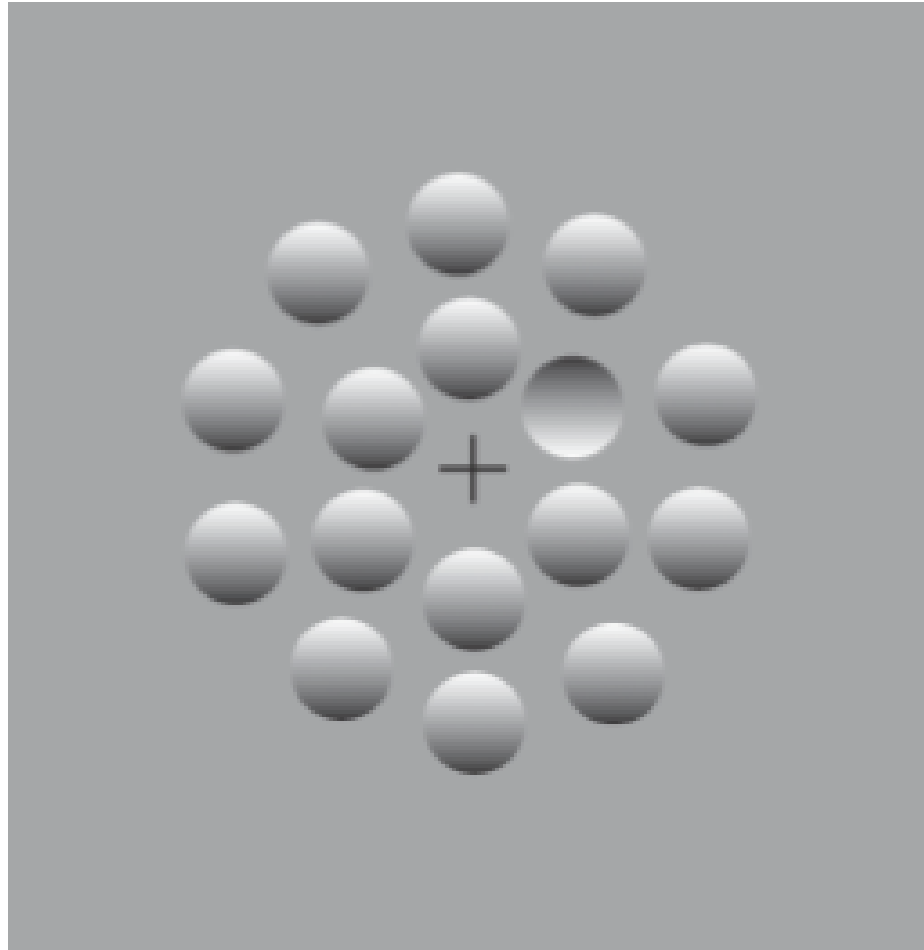
Basic Bayes and Discounting

What's the fuzz?!

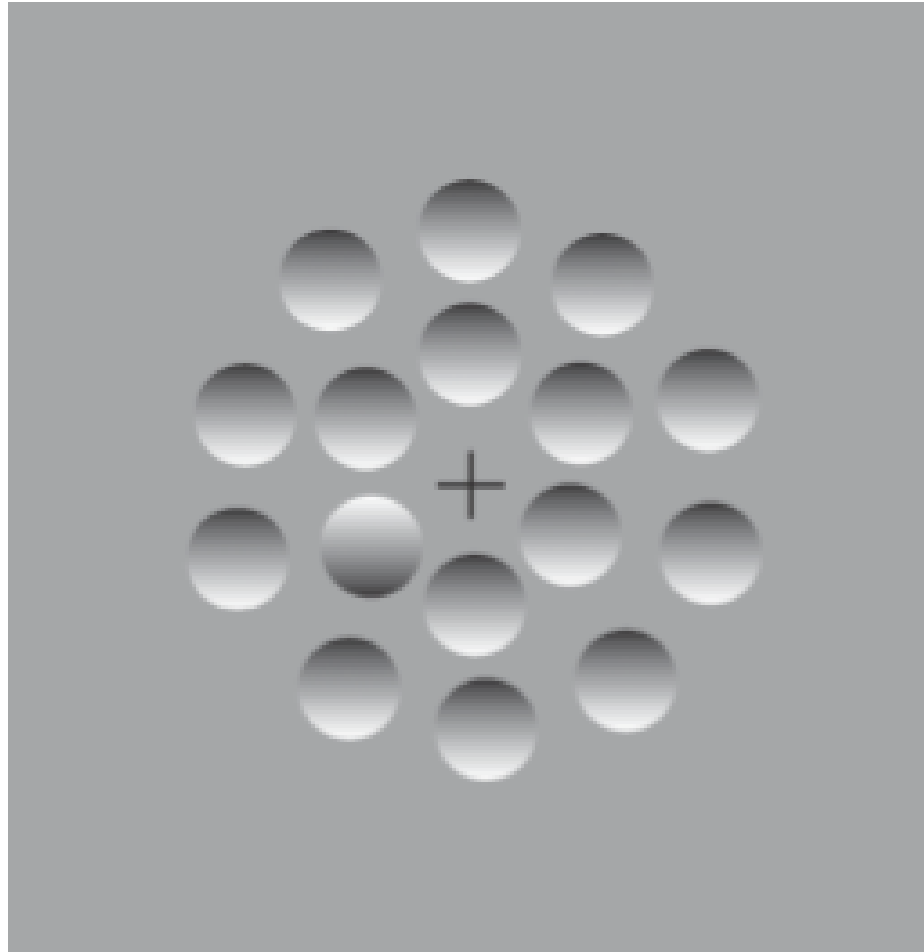
$$P(w \mid d) = \frac{P(d \mid w)P(w)}{P(d)}$$

- The problem of *any* observer
- The problem of perception
 - Probability of the world given sensory *data*
- Likelihood contains a model of the world
 - Not enough information
 - Prior adds information

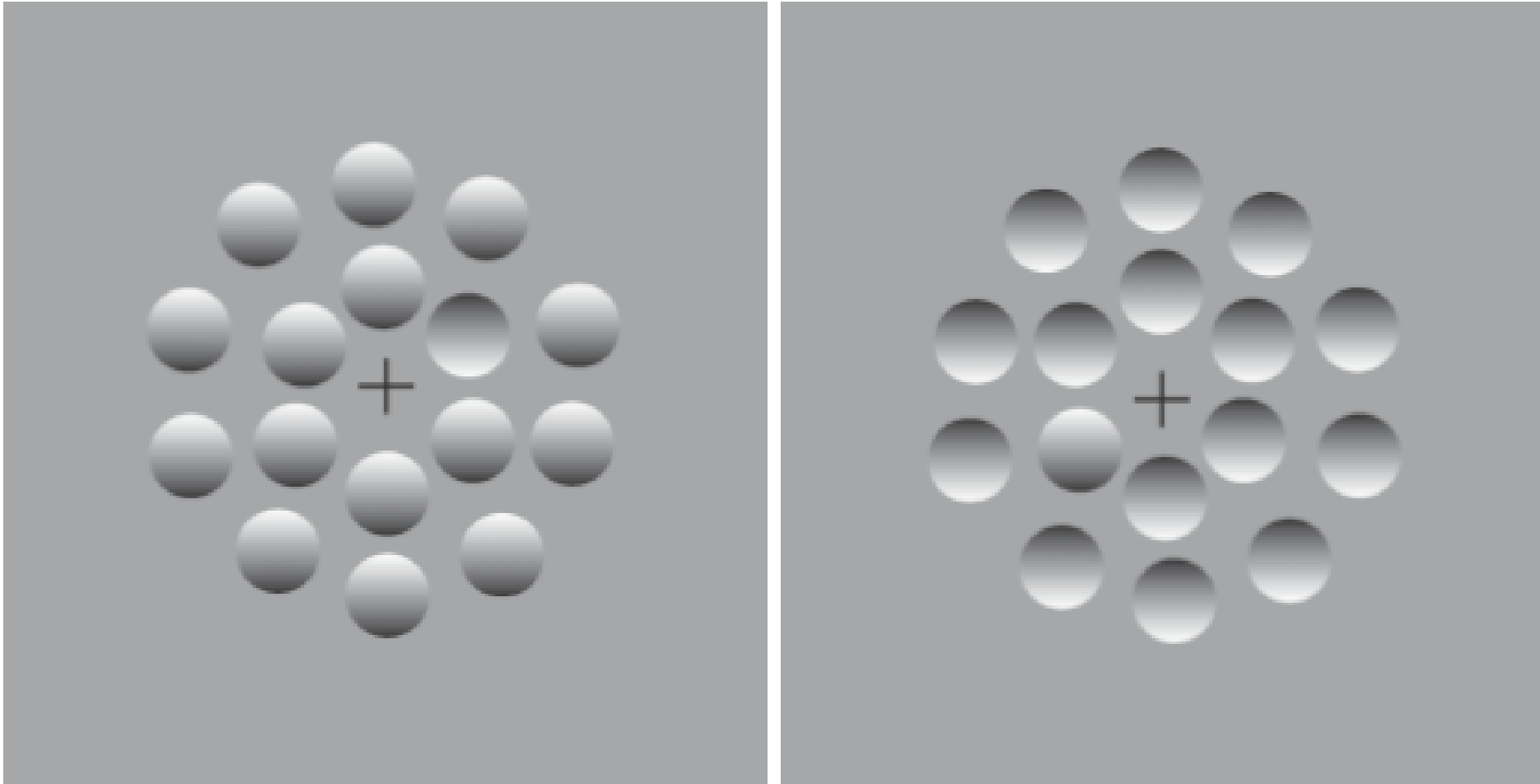
Protrusions or crevasses?



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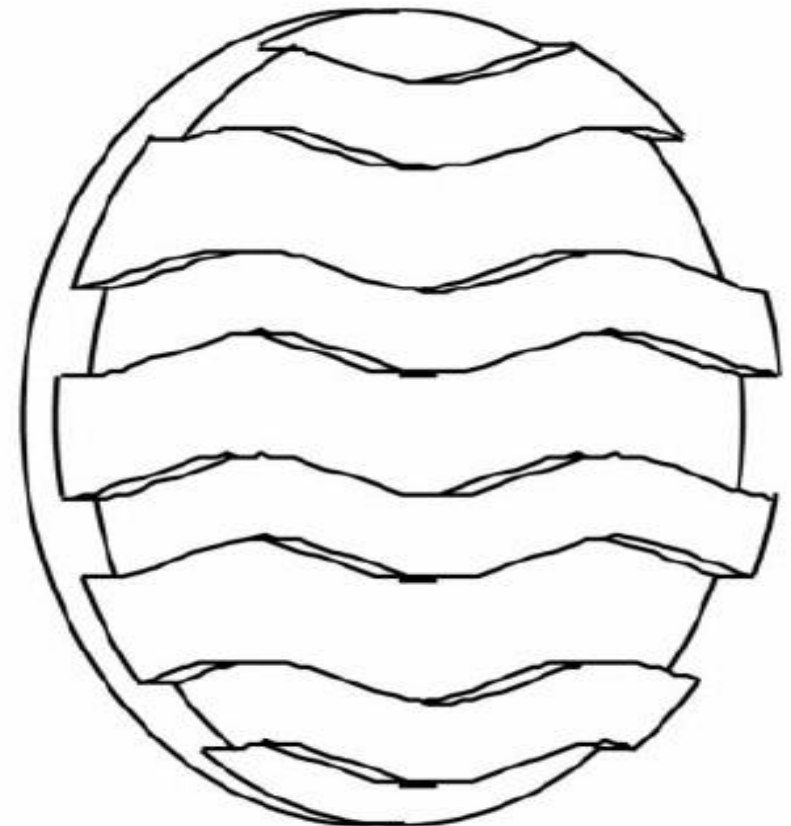
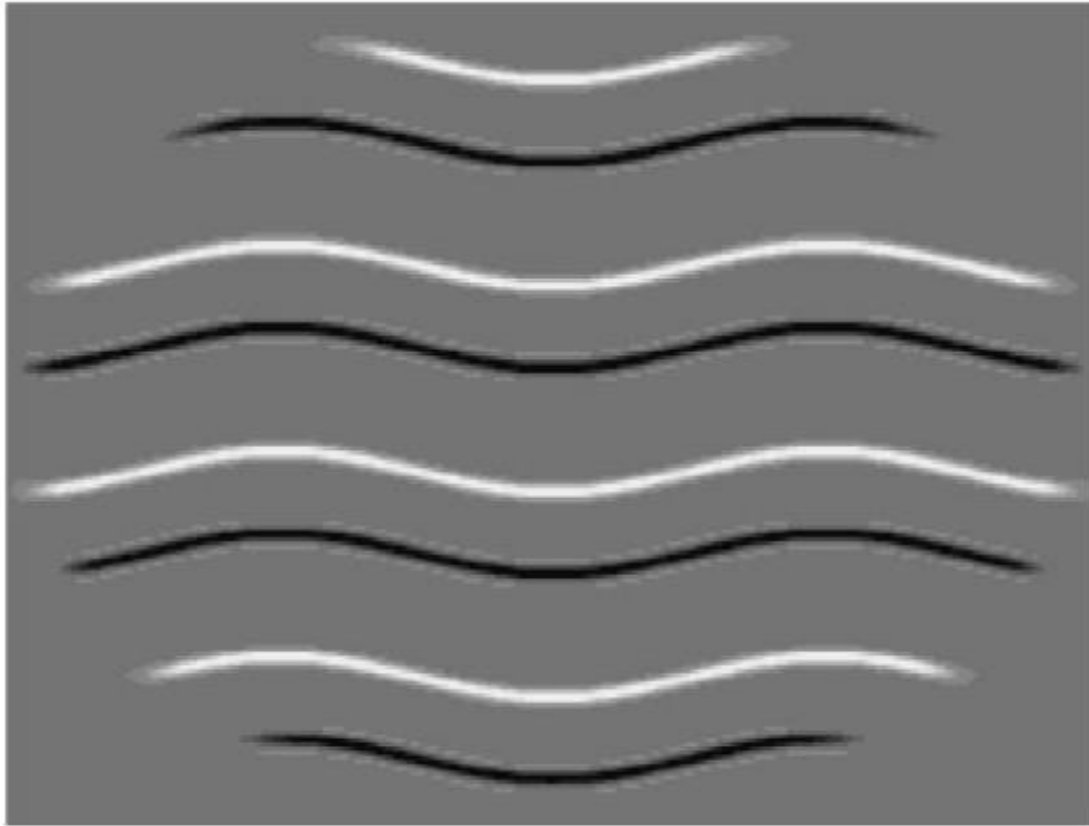


Light from above prior

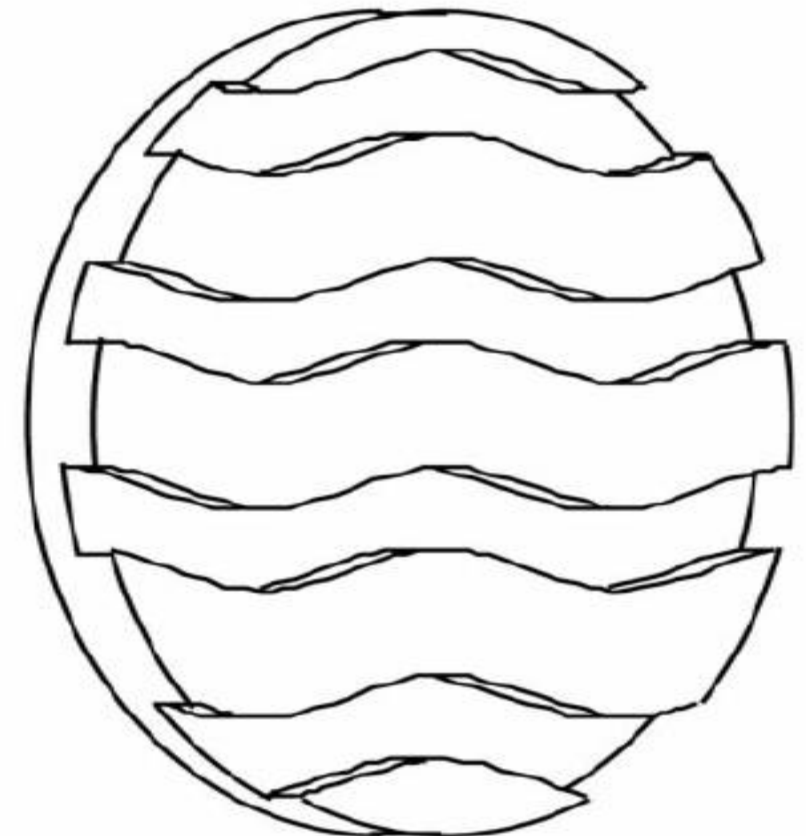
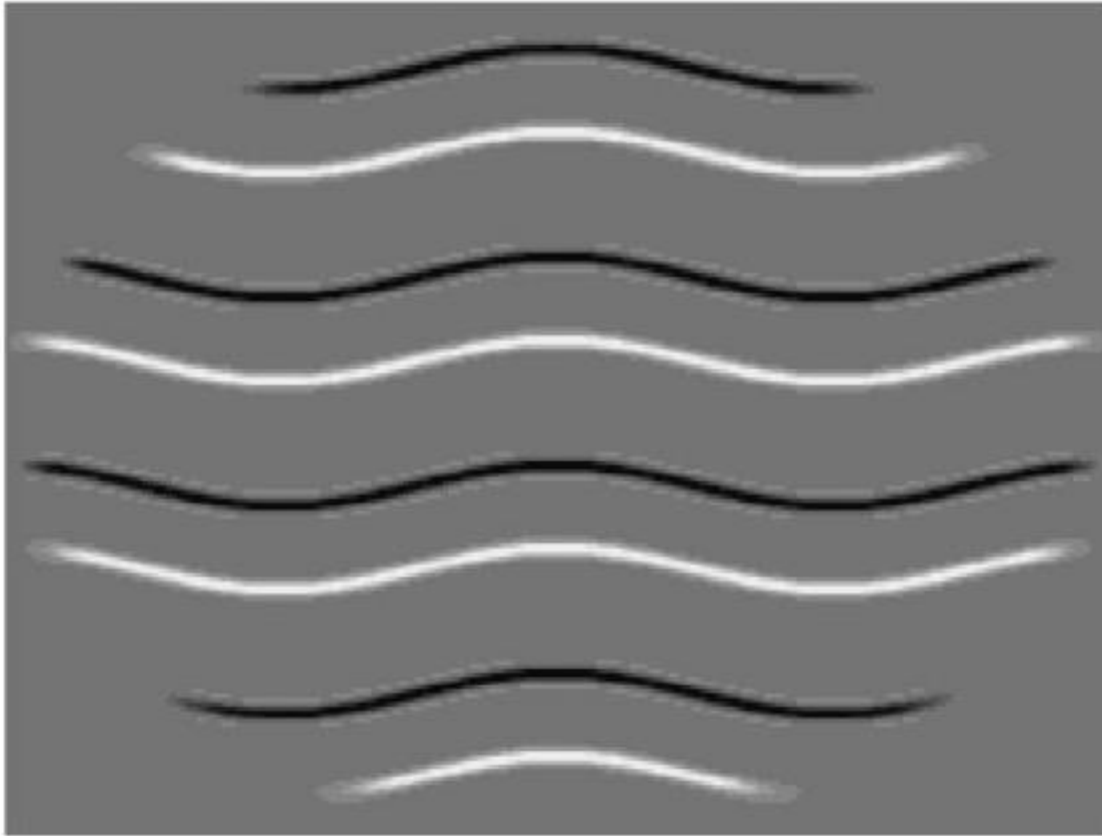
- The world, w , is 3-D shape
- ... but the prior is not on 3-D shape
- The prior is on the direction of the light source
- So how does that work with Bayes' rule?

$$P(w \mid d) = \frac{P(d \mid w)P(w)}{P(d)}$$

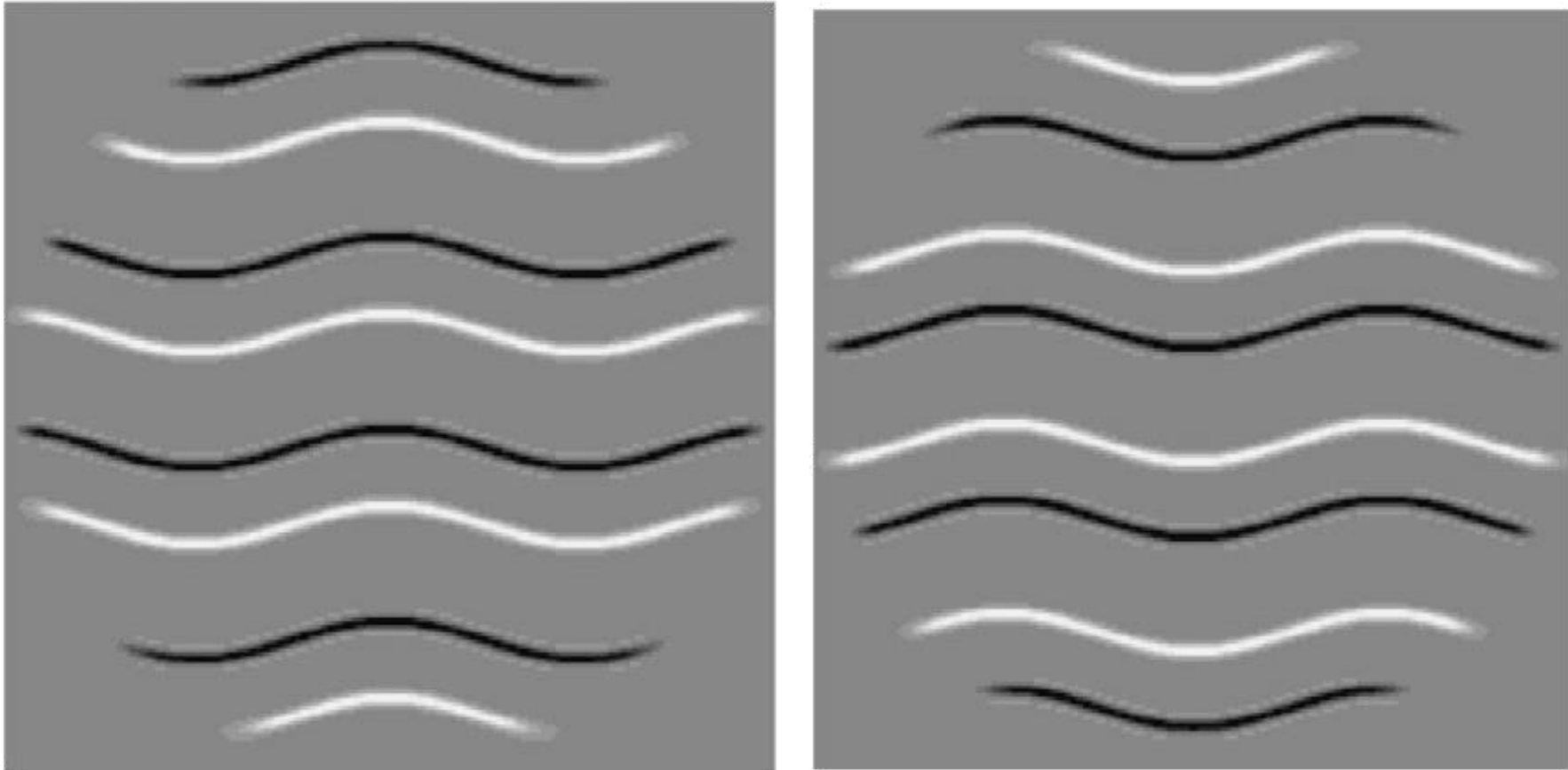
Example: Embossed surfaces



Example: Embossed surfaces



Example: Embossed surfaces



Marginalisation

- The task defines what is signal and what is noise
- Example: Find the bicycle
 - w_1 : presence of bicycle
 - w_2 : viewpoint
- Object constancy
 - Perceptual invariance



Marginalisation



Marginalisation

- We have a nuisance parameter
 - E.g. direction of the light source
- “Integrating out” - *marginalisation*
 - Or “summing out”
 - Is simply the weighted average of probabilities
 - Frees us from “unwanted” or “nuisance” parameters

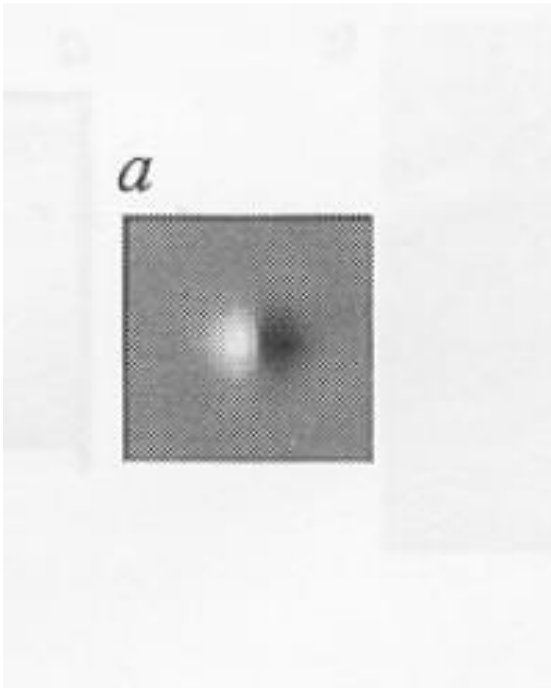
$$P(w_1 \mid d) = \sum_{w_2} P(w_1, w_2 \mid d)$$

Marginalisation

- What we got: $P(w_1, w_2 \mid d) = \frac{P(d \mid w_1, w_2)P(w_1, w_2)}{P(d)}$
- What we want: $P(w_1 \mid d) = \sum_{w_2} P(w_1, w_2 \mid d)$

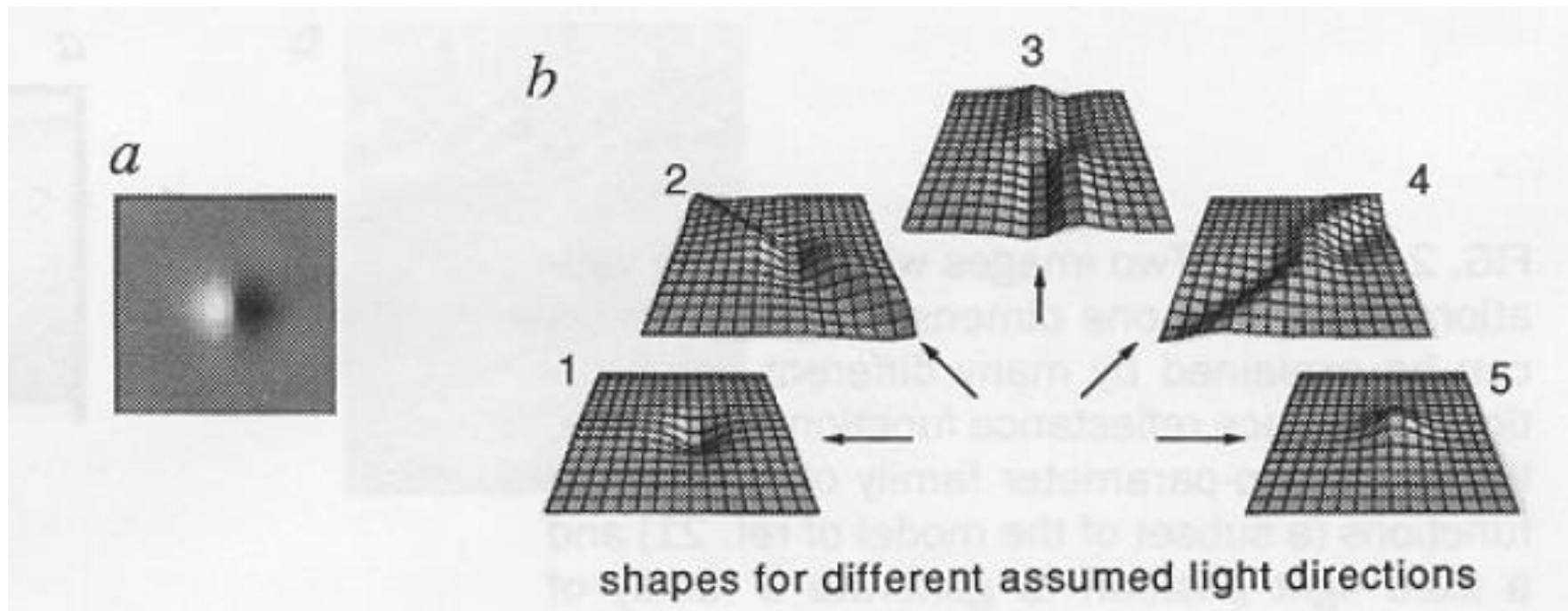
Marginalisation

- Example: Light from above prior



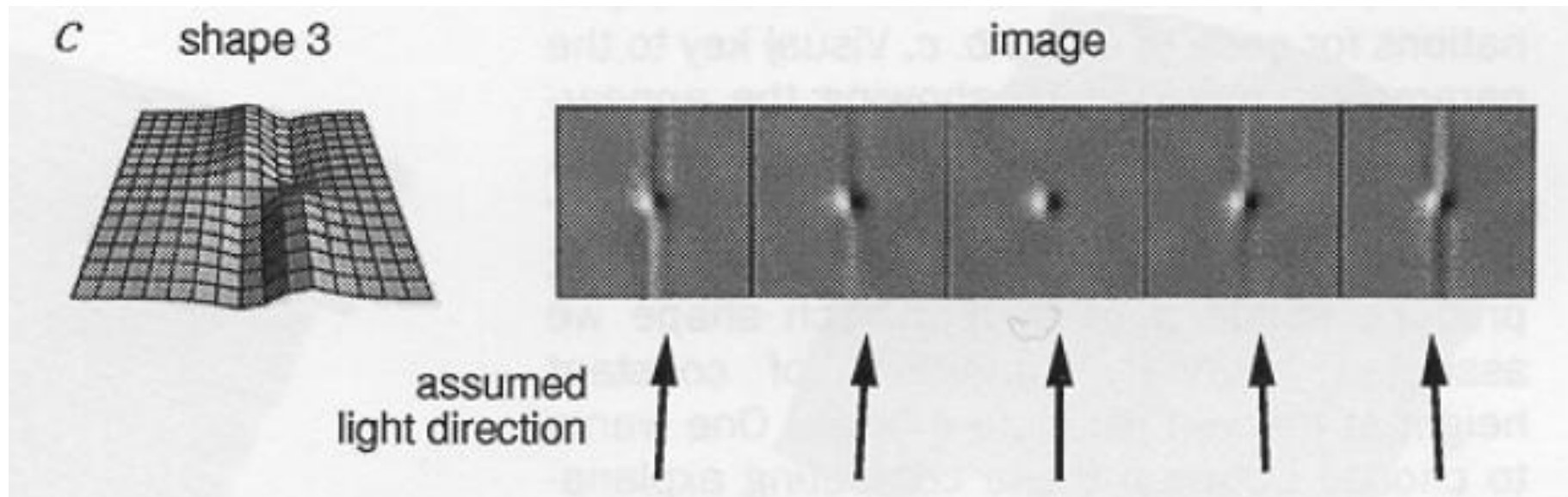
Marginalisation

- Example: Light from above (and slightly to the left?) prior
 - Weird shapes possible



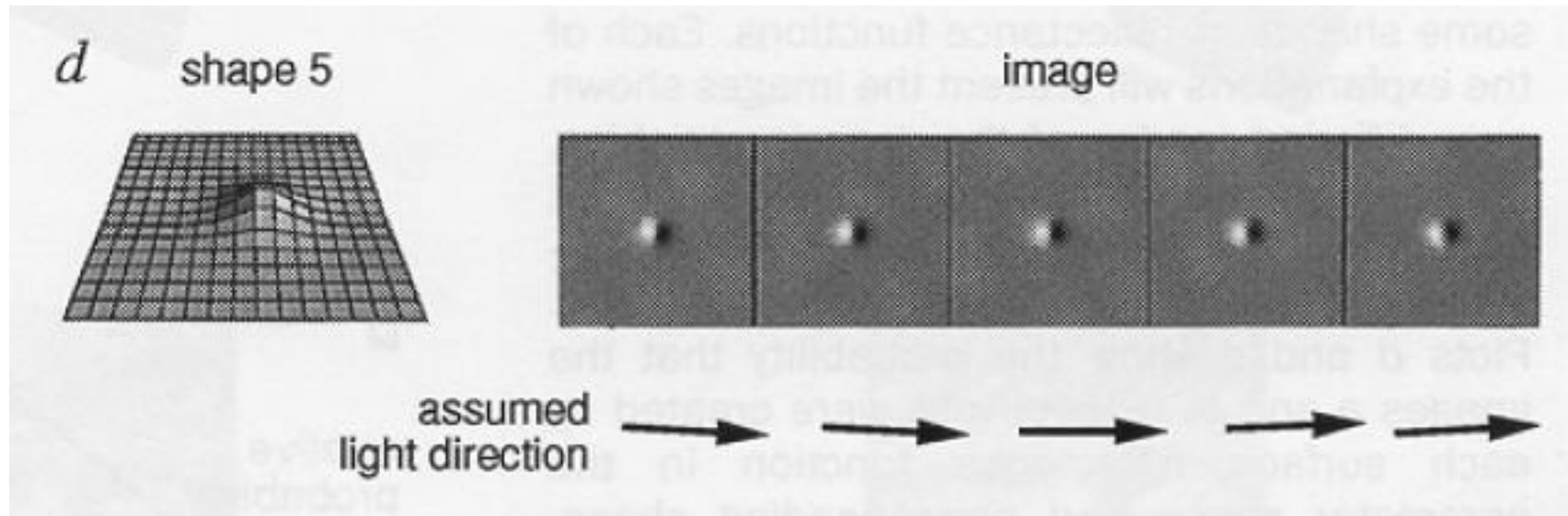
Marginalisation

- Weird shapes very sensitive to illuminant angle



Marginalisation

- Seen shapes not so sensitive to viewpoint



Marginalisation

- What we got:
$$P(w_1, w_2 \mid d) = \frac{P(d \mid w_1, w_2)P(w_1, w_2)}{P(d)}$$
- What we want:
$$P(w_1 \mid d) = \sum_{w_2} P(w_1, w_2 \mid d)$$
- Bayes factor
$$\frac{P(D \mid M_a)}{P(D \mid M_b)} = \frac{\int_{\theta_a} P(D \mid \theta_a, M_a)P(\theta_a \mid M_a)}{\int_{\theta_b} P(D \mid \theta_b, M_b)P(\theta_b \mid M_b)}$$



Marginalisation

- Interested in the shape of the object
- The viewing angle is a nuisance parameter
- The *generic viewpoint assumption*

Summary

- Perceptual unconscious inference
 - Solves ill-posed problems
 - by going beyond the information given
 - In Bayesian terms
 - Prior information
 - World model "hidden" in the likelihood
 - Searching for robust solutions by discounting nuisance effects
 - Summing or integrating over the components of the world, w , that are irrelevant for the task